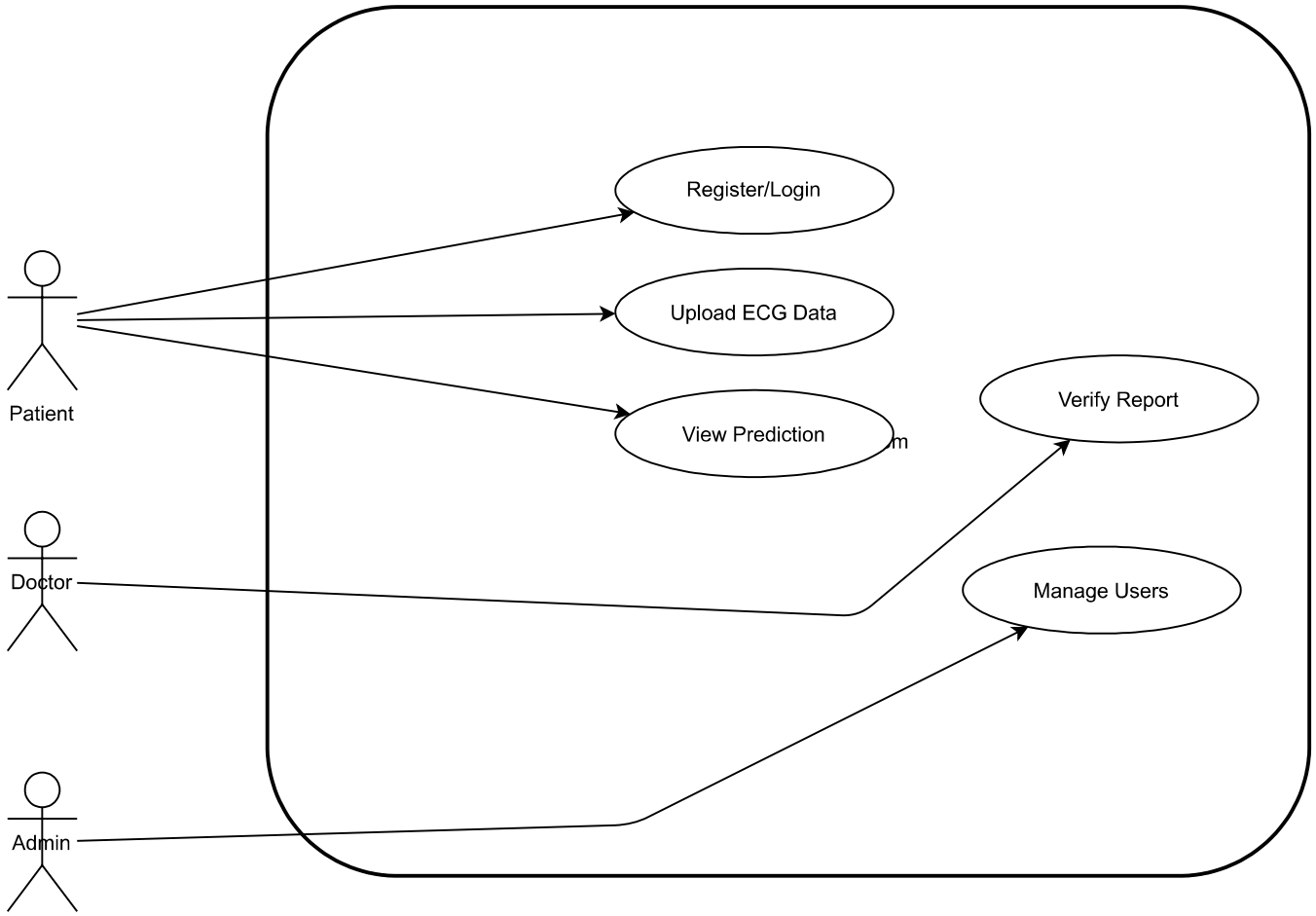
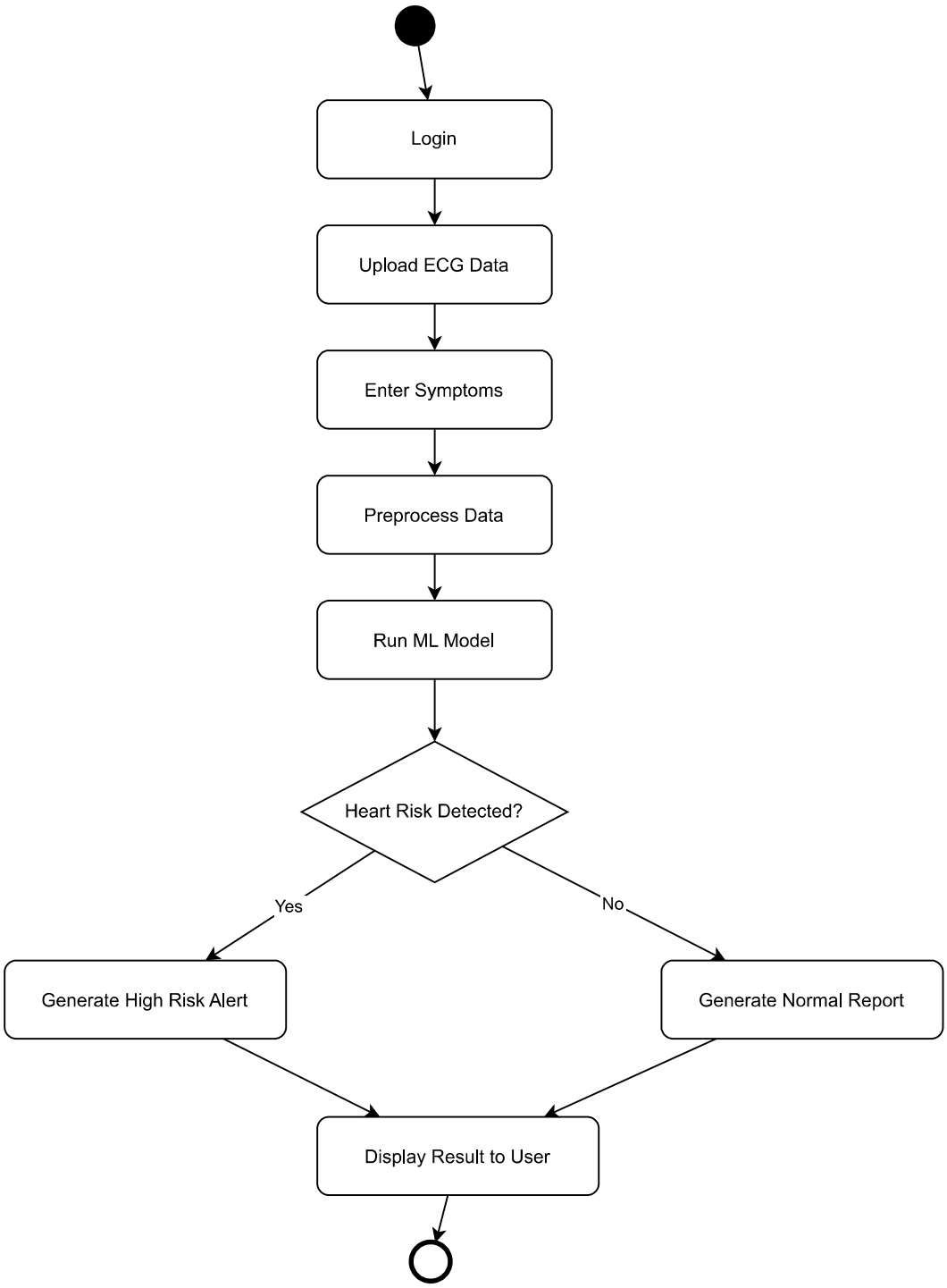
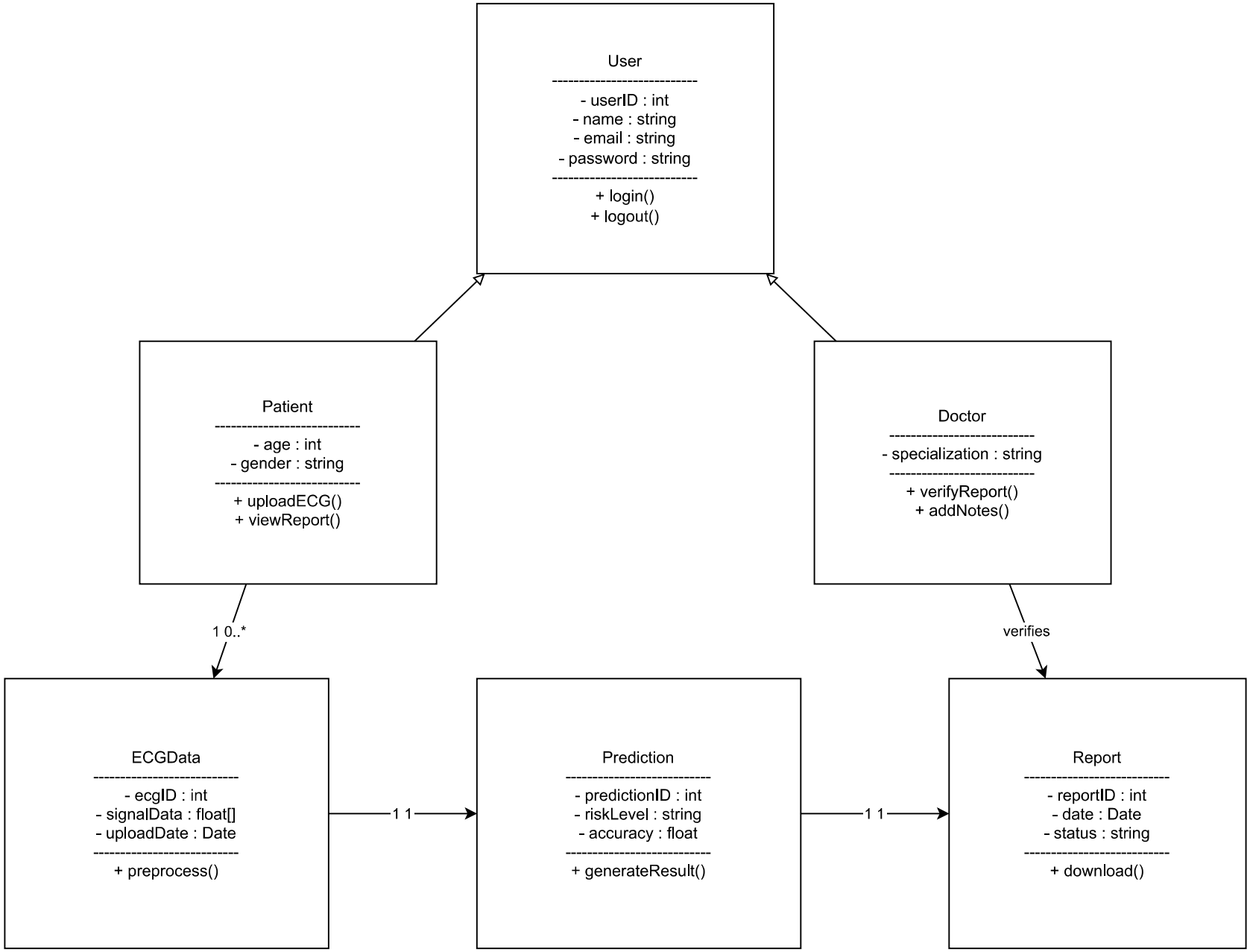
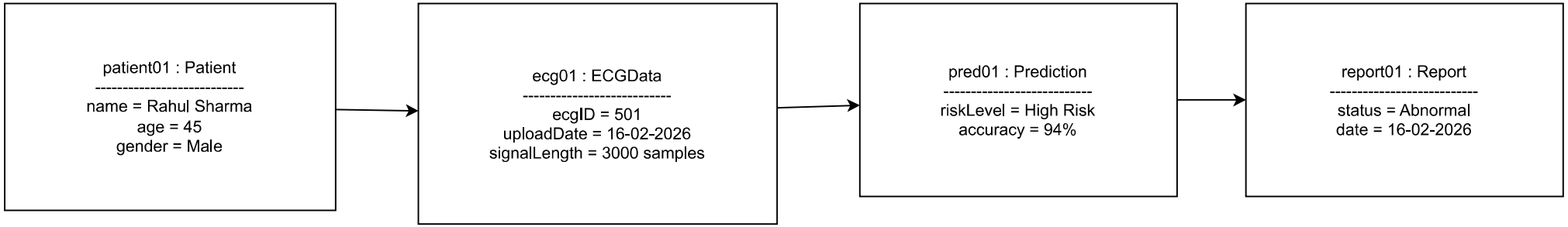
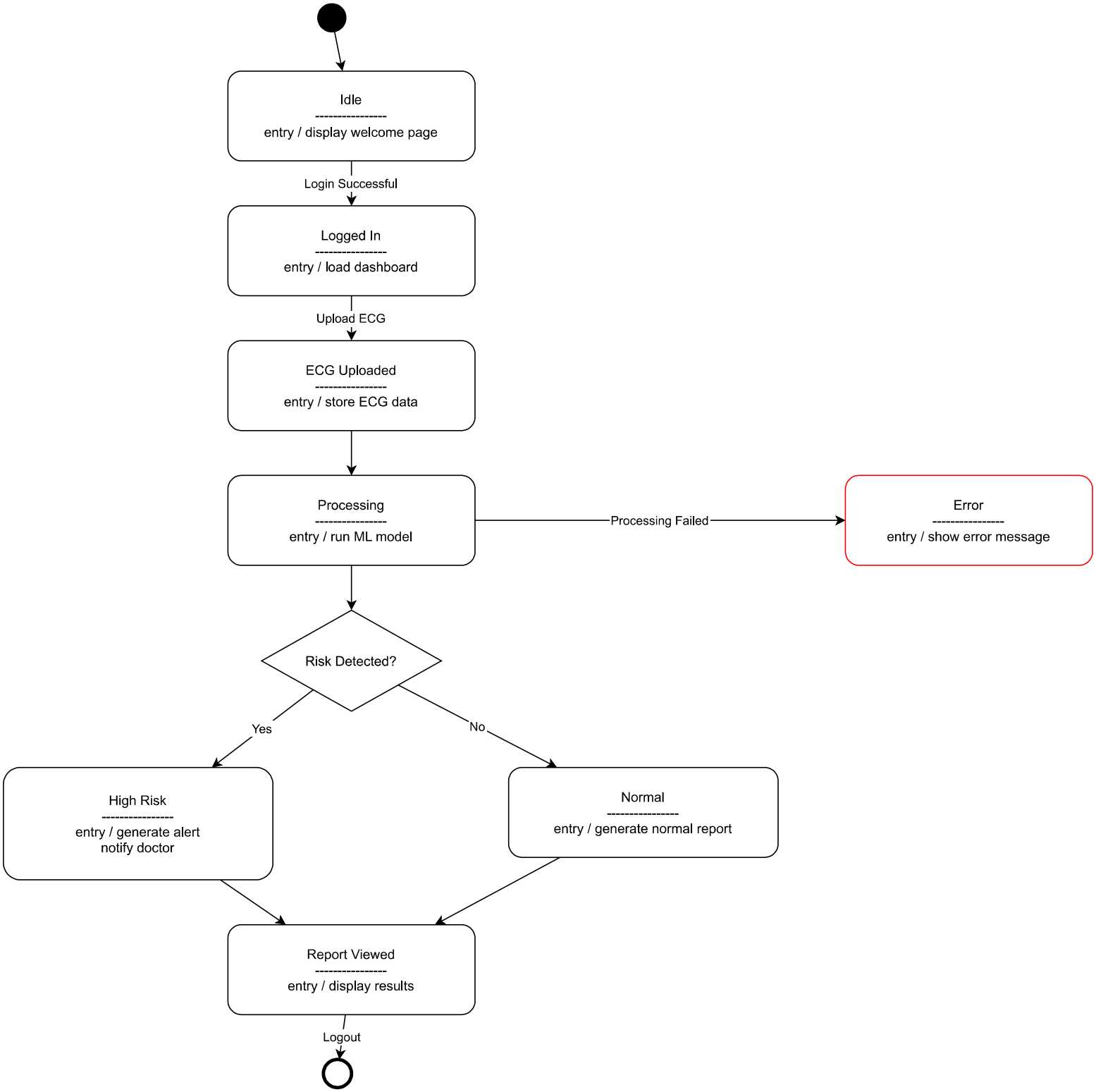

Register/Login
Upload ECG Data
View Prediction
Verify Report
Manage Users
Patient
Doctor
Admin


Login
Upload ECG Data
Enter Symptoms
Preprocess Data
Run ML Model
Heart Risk Detected?
Yes
No
Generate High Risk Alert
Generate Normal Report
Display Result to User


User
- userID : int
- name : string
- email : string
- password : string
+ login()
+ logout()
Patient
- age : int
- gender : string
+ uploadECG()
+ viewReport()
Doctor
- specialization : string
+ verifyReport()
+ addNotes()
1 0..*
verifies
ECGData
- ecgID : int
- signalData : float[]
- uploadDate : Date
+ preprocess()
1 1
Prediction
- predictionID : int
- riskLevel : string
- accuracy : float
+ generateResult()
1 1
Report
- reportID : int
- date : Date
- status : string
+ download()


patient01 : Patient
---
name = Rahul Sharma
age = 45
gender = Male
ecg01 : ECGData
---
ecgID = 501
uploadDate = 16-02-2026
signalLength = 3000 samples
pred01 : Prediction
---
riskLevel = High Risk
accuracy = 94%
report01 : Report
---
status = Abnormal
date = 16-02-2026


Idle
---
entry / display welcome page
Login Successful
Logged In
---
entry / load dashboard
Upload ECG
ECG Uploaded
---
entry / store ECG data
Processing
---
entry / run ML model
Processing Failed
Error
---
entry / show error message
Risk Detected?
Yes
No
High Risk
---
entry / generate alert
notify doctor
Normal
---
entry / generate normal report
Report Viewed
---
entry / display results
Logout